

VibeQueue

Lab 1 Version 1

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1. Introduction

1.1 Ecosystem Context

- Hospitality industry small cafes, coffee shops, and local restaurants.
- Atmosphere directly impacts customer satisfaction and dwell time

1.2 Problem Scope

- Over 1M+ independent food and beverage venues in the U.S. lack affordable, customer interactive music management tools
- Small businesses cannot justify high cost proprietary jukebox solutions

1.3 Current Solutions and limitations

- **TouchTunes:** proprietary jukebox hardware required; high upfront cost; designed for large bars and restaurants not accessible to small businesses [TouchTunes, n.d.]
- **Rockbot:** dedicated hardware recommended; medium cost; targets medium to large businesses; limited mobile/web interface [Rockbot, n.d.]
- **Manual staff requests:** customers verbally ask staff to change songs; inefficient use of employee time during peak hours; no consistency
- **Generic personal streaming (e.g., Spotify Free/Personal):** illegal for commercial use without commercial license; no customer interaction feature

1.4 Market Gap

- No real time solution that is:
 - Affordable
 - Real-time
 - Staff free
 - Hardware free
 - Customer interactive
 - Legally compliant music management solution exists for small local venues
- Customers have no way to influence venue atmosphere digitally
- Owners missing potential secondary revenue stream from music engagement

1.5 Required Solution Characteristics

- Low cost must be viable on a small business budget
- No proprietary hardware requirement
- Customer facing interaction without interrupting staff
- Legal commercial music licensing compliance (ASCAP, BMI, SESAC)\

- Owner/staff control over final song selection to protect venue atmosphere
 - [Figure 1: Current Process Flow customer has song request > staff intervention > decision: capable of playing specific song? > decision: acceptable to business? > song played or not played]
 - Introduce VibeQueue as solution: a web based, hardware agnostic, subscription jukebox platform for local spaces tagline: “Your Vibes, Managed”

2. VibeQueue Product Description

2.1 Overview

- Top level description: VibeQueue is a software only solution that replaces expensive proprietary hardware and manual staff processes with a seamless digital music ecosystem for small local spaces

2.2 Goals and objectives:

- Give customers a frictionless way to influence the venue’s music via their own smartphones
- Give owners and staff complete control over which songs can be requested
- Provide an optional secondary revenue stream (pay to request model)
- Ensure all music playback is legally licensed for commercial use
- Be deployable with zero new hardware purchase

2.3 Unique differentiators vs. competition:

- Software only runs on devices the business already owns
- Subscription based low monthly cost vs. high upfront hardware investment
- Customer facing QR interface vs. staff mediated requests
- Legally compliant via SoundMachine (ASCAP/BMI/SESAC) vs. unlicensed personal streaming

2.4 System Flow

- [Figure 2: VibeQueue Solution Process Flow (MFCD) customer scans QR > selects song in web app > Vibe Engine checks genre/constraints against Venue Database > validated song sent to Local Server > SoundMachine plays audio]

2.5 Key Product Features and Capabilities

2.5.1 QR Code Access

- Customers scan a QR code on their table to access the venue’s VibeQueue portal in a mobile browser no app download required; QR links customer device to the business’s specific portal

2.5.2 Song Browsing & Requests:

- Customers search and browse songs using metadata provided by the Soundcharts API; results filtered in real time based on venue's active vibe constraints

2.5.3 Vibe Engine (Custom Song Selection):

- proprietary matching algorithm built by the VibeQueue team; evaluates songs against venue's allowed genre, tempo, mood, popularity settings; rejects non matching requests before they reach the queue
 - Significant because it replaces the need for the Spotify API (which has strict commercial use restrictions) with a legally safe, in house solution
 - Allows venue to maintain brand atmosphere even with customer input

2.5.4 Business Control Center:

- owner/staff facing dashboard the core management interface
 - Set genre allowlists and blocklists (e.g., "acoustic only," "no hip hop")
 - Block explicit content automatically
 - Manage live song queue (skip, reorder, remove)
 - Manage user sessions (ban abusive users)
 - View analytics: most requested songs, peak request times

2.5.5 Dynamic Vibe Moderation:

- integrated content filters applied system wide; owners configure tempo range, mood tags, and explicit content rules once; system enforces automatically on every request

2.5.6 Pay to Request:

- optional monetization feature; customers pay a small fee (e.g., \$0.99–\$1.99) to boost a song to the top of the queue; automated payment processing
 - Revenue split: business keeps majority; VibeQueue takes a 5% platform cut to sustain development
 - Unique no competitor offers customer facing paid interaction at this price point for small businesses

2.5.7 Licensed Streaming via SoundMachine:

- all audio playback routed through SoundMachine, which holds commercial music licenses (ASCAP, BMI, SESAC); ensures VibeQueue is fully legal for business use

2.5.8 Hardware Agnostic Deployment:

- runs on any internet connected device the business already owns staff phones, existing tablets, a dedicated laptop; no proprietary console purchase required

2.6 Major Components (Hardware/Software)

2.6.1 Hardware requirements:

- No proprietary or dedicated hardware required
- Runs on any internet connected device (tablet, smartphone, laptop/desktop) the venue already owns
- Venue speaker system connected to the device running SoundMachine (existing audio setup)

2.6.2 Software architecture (ref: MFCD diagram, Figure 2):

- **Web App UI:** customer facing interface accessed via QR code; built as a responsive web app; search interface connected to Soundcharts API
- **Vibe Engine:** custom built song selection and validation logic; receives customer song request > queries Soundcharts for metadata > checks against Venue Database constraints > returns validated/rejected result
- **Soundcharts API:** third party music analytics service; provides commercially safe metadata (genre, tempo, mood, popularity, explicit flag) for song lookup and vibe matching
- **SoundMachine:** third party licensed commercial streaming service; handles actual audio playback in the venue; ensures ASCAP/BMI/SESAC compliance
- **Local Server:** handles audio signal routing, playback status feedback, and communication between Vibe Engine and SoundMachine
- **Venue Database:** stores venue specific vibe settings (genre rules, explicit flags, tempo limits), song queue state, and transaction/revenue data
- **Business Control Center Dashboard:** owner/staff interface for real time queue management, constraint configuration, and analytics viewing

2.6.3 Technologies, APIs, and frameworks:

- Tech stack: LAMP (Linux, Apache, MySQL, PHP)
- Soundcharts API music metadata
- SoundMachine API licensed audio playback
- QR code generation library (venue specific portal linking)
- Web based front end (mobile first responsive design)
- Architecture note: Initial prototype explored Spotify API; pivoted due to strict Spotify commercial use licensing restrictions; current architecture uses Soundcharts + SoundMachine to eliminate legal risk

3. Identification of Case Study

- 3.1 WHO

- “Brewpoint Café” single location independent coffee shop; owner (Jamie) + 2 barista staff; ~30 seats; primarily serves morning commuters and remote workers 7 AM–2 PM
- 3.2 WHAT PROBLEM
 - Baristas are interrupted 10–15 times per morning rush by verbal song requests; owner loses control of the café’s intended “acoustic/lo fi” vibe when staff accommodate requests; no revenue is generated from music engagement
- 3.3 WHICH FEATURES TESTED
 - QR Code Access (on each table card)
 - Song Browsing & Requests (Soundcharts powered search)
 - Business Control Center (genre restrictions: acoustic, indie, lo fi only; explicit content blocked; max tempo 120 BPM)
 - Dynamic Vibe Moderation (auto enforcement of genre and tempo rules)
 - Pay to Request (\$0.99 boost option during morning rush)
- 3.4 HOW (Scenario):
 - Customer “Alex” sits down, notices QR card on table, scans with phone
 - VibeQueue portal opens in browser no download needed; Alex sees current queue and search bar
 - Alex searches “Golden Hour” by JVKE; Vibe Engine checks: genre = pop/indie (allowed), explicit = no (pass), tempo = 98 BPM (under 120 BPM limit, pass) > song added to queue
 - Alex optionally pays \$0.99 to boost song to top of queue; payment processed automatically
 - Staff receive zero interruption; song plays through SoundMachine on café speakers
 - Owner Jamie logs into Control Center on the counter tablet during slow period; reviews queue, sees request analytics, adjusts genre allowlist for afternoon shift
 - [Figure 3: Case Study scenario flow Alex scans QR > searches song > Vibe Engine validates > song queued > SoundMachine plays > staff uninterrupted]
- 3.5 FUTURE EXPANSION:
 - Scale to small cafe chains (5–10 locations) using centralized Control Center with per location vibe profiles
 - Expand to adjacent markets: college dining halls, hotel lobbies, waiting rooms, fitness studios, barbershops
 - Potential B2B SaaS licensing model for venue management companies
 - Add loyalty/gamification features: customers earn “vibe points” for requests that stay in the queue

4. Glossary

- **API (Application Programming Interface):** a set of rules and protocols that allows different software applications to communicate with each other
- **ASCAP (American Society of Composers, Authors, and Publishers):** a U.S. performing rights organization that licenses music for commercial use
- **BMI (Broadcast Music Inc.):** a U.S. performing rights organization that manages commercial music licensing on behalf of songwriters and publishers
- **Business Control Center:** VibeQueue's owner facing web dashboard for configuring vibe constraints, managing the song queue, and viewing analytics
- **Hardware Agnostic:** software designed to run on any existing device without requiring specific or proprietary hardware
- **LAMP:** an open source web development stack comprised of Linux (OS), Apache (web server), MySQL (database), and PHP (scripting language)
- **MFCD (Major Functional Component Diagram):** a visual representation of a system's major components and the data flows between them, as defined in CS 410
- **QR Code (Quick Response Code):** a type of two dimensional matrix barcode that can be scanned by a smartphone camera to open a URL or application
- **SESAC (Society of European Stage Authors and Composers):** a U.S. performing rights organization providing commercial music licensing
- **SoundMachine:** a third party commercial music streaming service that holds ASCAP/BMI/SESAC licenses; used by VibeQueue for legal audio playback in venues
- **Soundcharts API:** a third party music analytics API that provides commercially safe song metadata including genre, tempo, mood, popularity, and explicit content flags
- **Vibe Engine:** VibeQueue's proprietary, in house song selection and validation algorithm; evaluates customer song requests against venue's configured constraints before allowing them into the queue

5. References

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